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FEDERAL SPECIFICATION

LADDERS, STRAIGHT, EXTENSION, AND TRESTLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers straight, extension, trestle and extension trestle ladders and safety shoes for these ladders.

1.2 Classification.

1.2.1 Types, classes, grades, and kinds. Ladders shall be of the following types, classes, grades, and kinds as specified (see 6.2):

Types and classes:

Type I - Single, straight.
Type II - Extension, straight.
Class a - Two section.
Class b - Three section.
Type III - Trestle.
Type IV - Extension, trestle.

Grades:

Grade A - Industrial, heavy duty.
Grade B - Commercial, medium duty.
Grade C - Household, light duty.

Kinds:

Kind 1 - Aluminum.
Kind 2 - Magnesium (types I and II only).

1.2.2 Size. Ladders shall be of the sizes listed in tables I and II, as specified (see 6.2).

TABLE I. Size, types I and II ladders, kinds 1 and 2

Type	Class	Grade	Size in feet 1/
I	-	A	6 to 30
I	-	B	6 to 24
I	-	C	6 to 20
II	a	A	16 to 48
II	a	B	16 to 44
II	a	C	16 to 40
II	b	A, B	40 to 72

1/ The size designation for type II ladders is the industry designation for size, it is not the working length; to determine the working length subtract the overlap, as indicated in table IV, from the size in table I.

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TABLE II. Size, types III and IV ladders, kind one only

		Size in feet	
Type	Grade	Basic	Extended
III	A,B	6 to 20	--
IV	A,B	6	9
		8	12
		10	16
		12	20
		14	24
		16	28
		18	32

1.2.3 Safety shoes. Safety shoes for all types of ladders shall be of the following styles, as specified (see 6.2):

S1 - Flat pad, swivel.
S2 - Flat pad, non-swivel.
S3 - Vacuum cup.
S4 - Ice pick.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Steel Strapping, Flat.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-F-320 - Fiberboard; Corrugated and Solid Sheet Stock (Container Grade) Open and cut Shapes.
PPP-S-760 - Strapping, Nonmetallic, (and Connectors).
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (**Civil Agencies**).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 Material. Kind 1 ladders shall be fabricated from aluminum alloys 6061 or 6063 T6 temper. Kind 2 ladders shall be fabricated from magnesium of any alloy and temper which meets all provisions of this specification.

3.1.2 Flare. Ladders shall have parallel side rails or have side rails varying uniformily in separation along the length (tapered) or have side rails flaring at the base, at the manufacturer's option.

3.1.3 Side rails. Shape and size of side rails shall be such as to insure that ladders conform to the requirements of this specification.

3.1.4 Rungs and connections. Rungs and connections shall be constructed to assure rigidity as well as strength. Rungs shall not be omitted at the lock position of fly sections of type II, grade A or B ladders unless the lock is located at the bottom end of the fly section, or the fly section is mounted on the underneath side of the base section.

3.1.4.1 Spacing. The spacing between rungs, measured along the side rails shall be 12 ± 1/8 inches on centers.

3.1.4.2 Surface. Rungs shall be corrugated, knurled, or dimpled to minimize slipping.

3.1.5 Extension locking device, types II and IV. Types II and IV ladders shall be equipped with extension locking devices. The locking devices shall not show any permanent deformation when loaded (see 4.4.1.1).

3.1.6 Welds. The surfaces of parts to be welded shall be free from paint, grease and dirt. Welds shall develop adequate strength and transmit stress to parts connected under proof and service loading without failure. Welds shall be deposited by the inert gas (argon, helium) shielded-arc method using filler wire compatible with the base metal. Components shall be positioned and secured to assure proper alignment.

3.1.7 Hardware, including fasteners. Hardware shall meet the strength requirements of the ladders' component parts, and shall be corrosion-resistant or protected against corrosion. Hardware shall be made of material that is compatible with the base metal (see 4.4.1.1).

3.1.8 Nonslip shoes. Unless otherwise specified (see 6.2), ladders shall be equipped with a pair of safety shoes. Nonslip shoes shall be capable of sustaining the loads imposed upon them and shall be securely attached to the side rails in such a manner as not to weaken the side rails (see 4.4.1.1).

3.1.8.1 Flat pad, swivel, S1, (type I and II ladders only). Swivel shoes shall be constructed of metal and shall be attached to the base end of the ladder side rails. Shoes shall have non-slip pads. The minimum area of each non-slip pad shall be in accordance with table III. Shoes shall remain parallel to the floor surface regardless of the angle of the ladder.

3.1.8.2 Flat pad, non-swivel, S2, (type III and IV ladders only. Non-swivel shoes shall be constructed of metal and shall be attached to the ends of side rails in such a manner as not to weaken the side rails. Shoes shall have non-slip pads securely attached to their metal frames. The minimum area of each non-slip pad shall be in accordance with table III. Shoes shall remain parallel to the floor surface when the ladder is erected at the normal working angle of 75-1/2°.

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TABLE III. Pad area for S1 and S2 safety shoes, square inches

Ladder grade	Pad area, minimum
A	8
B	6
C	4

3.1.8.3 Vacuum cup, S3. Vacuum cup shoes shall be constructed of non-slip material and shall be attached by ball and socket to a stirrup-type holder. The minimum area of each vacuum cup shall be 10 square inches for all three grades of ladders. Ball and socket shall allow the vaccum cup shoe to fit the contour of the surface regardless of the angle of the ladder. Stirrups shall be cast of malleable or ductile iron, steel or aluminum, or they shall be stamped from aluminum or steel, and they shall be securely attached.

3.1.8.4 Ice picks, S4. Ice picks shall be constructed of metal and either be attached to the ends of the ladder side-rails, or be an integral part of the safety shoes. Ice picks shall be in the form of ice-pick plates, hardened spikes, or prongs.

3.1.9 Leveling device. When specified (see 6.2), a leveling device shall be provided for use with ladders set up on uneven surfaces such as sloping driveways, steps, or uneven ground. The device shall be principally of metal construction and shall be equipped with safety shoes which have non-slip pads. The area of each pad shall be in accordance with table III. Ice picks shall be provided when specified (see 3.1.8.4 and 6.2) in addition to, or in lieu of safety shoes. The leveling device shall be adjustable for height variations by manual or automatic means. The leveling device shall be capable of sustaining the loads imposed upon it (see 4.4.1.1).

3.2 Types I and II, single and extension ladders.

3.2.1 Width. The minimum clear width between side rails of single straight, or any section of an extension ladder shall be 12 inches.

3.2.2 Length tolerance. A tolerance of plus or minus 1/2 inch is permissible for length of each section.

3.2.3 Overlap, type II. Each section of multisection ladders, when fully extended, shall overlap the adjacent section in accordance with table IV. Tolerance of the overlap shall be plus or minus 1 inch for a two-section ladder and plus or minus 2 inches for three-section ladders.

TABLE IV. Overlap, extension ladders

Length of ladder	Two-section Overlap	Three-section Overlap
feet	feet	feet
Up to 36 inclusive	3	6
Over 36 to 48 inclusive	4	8
Over 48	5	10

3.2.4 Overlap control, type II. Extension ladders shall be equipped with a positive means to prevent ladders from being used beyond the overlap shown in table IV.

3.2.5 Rope and pulley, type II. Extension ladders shall be equipped with a rope and pulley attached to the ladders in such a manner as not to weaken either rungs or the side rail. Pitch diameter of the pulley shall not be less than 1-inch. The rope used with the pulley shall be of sufficient strength to raise the weight of the fly section or sections with a safety factor of 10, and shall be of sufficient length to extend the ladder to its full working length.

3.2.6 Hooks. When specified (see 6.2), hooks attached at or near the end of ladders shall be furnished for added security. Hooks shall be of all metal construction and shall be securely bolted or riveted to the side rails or affixed to the ladder by equivalent construction. Hooks shall be capable of sustaining loads imposed upon them (see 4.4.1.1).

3.2.7 Caps. The top of straight single ladders, and the top of the upper fly section of extension ladders shall be equippped with caps. Caps shall be cast aluminum, formed aluminum strip integrally formed ends, plastic, or corrosion-resistant coating on steel and covered with rubber. Plastic shall be melt mixed vinyl (see 4.4.2.1) or acrylonitrile-butadiene-styrene (ABS).

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3.2.8 Weight. The weight of the type I and II ladders shall be at the option of the manufacturer, unless otherwise specified (see 6.2).

3.3 Types III and IV, trestle and trestle extension ladders.

3.3.1 Length. Trestle ladders, extension sections, or base sections of extension trestle ladders shall be not more than 20 feet in length. A tolerance of plus or minus 1/2 inch is permissible for the length of each section.

3.3.2 Width. The minimum clear distance between side rails of the trestle or extension section at the narrowest location shall not be less than 12-1/2 inches. The width flare shall not be less than 1 inch per foot of length of side rail.

3.3.3 Spread. With section opened, the spread of base shall not be less than 5-1/2 inches per foot of base section side rail.

3.3.4 Spreaders. Metal spreaders and locking devices shall be of sufficient strength to securely hold the front and back sections in the open position, and shall be a component part of each trestle ladder. Height of spreaders shall not be greater than 2/3 the height of the ladder.

3.3.5 Weight. The weight of the type III and IV ladders shall be at the option of the manufacturer, unless otherwise specified (see 6.2).

3.4 Strength requirements.

3.4.1 Types I, II, III and IV.

3.4.1.1 Rung strength.

3.4.1.1.1 Bending. Rungs shall withstand loading without failure or permanent deformation in excess of 1/100 of the clear span between the inside webs of the side rails when tested in accordance with 4.4.1.2.1.

3.4.1.1.2 Rung to side rail shear. The fastening means for attaching rungs to the side rails shall not fail when tested in accordance with 4.4.1.2.2.

3.4.1.1.3 Torque. The alternating torque load shall cause no visible relative motion between rung and side rails when tested in accordance with 4.4.1.2.3.

3.4.1.2 Side rail cantilver bending. Rails shall return to their original position without any visible defects or permanent deformation when tested in accordance with 4.4.1.2.4.

3.4.2 Types I and II, single and extension ladders.

3.4.2.1 Horizontal bending. When tested in accordance with 4.4.2.2 there shall be no permanent deformation in excess of 1/1000 of the effective span of the ladder side rails and there shall be no other weakening or failure of the structure when loaded.

3.4.2.2 Side sway. Deflection of the mid-point of the lower side rail of the loaded ladder shall not exceed the value in table V, when tested in accordance with 4.4.2.3. On removal of the load, the ladders shall return to their original position without permanent deformation is excess of 1/1000 of the effective span of the side rails.

TABLE V. Side sway deflection

Test section length	Mid-span deflection of lower side rail, maximum
feet	inches
8	1
10	1-1/4
12	1-1/2
14	1-3/4
16	2
18	2-1/4
20	2-1/2
22	2-3/4
24	3

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3.4.2.3 Deflection. The deflection of the loaded rail, and the difference in deflection between loaded and unloaded rails shall be in accordance with table VI (see 4.4.2.4).

TABLE VI. Deflection

Length of ladder	Deflection of loaded rail, maximun	Difference in deflection between loaded and unloaded rails measured from horizontal, maximum
feet	inches	degrees
16	2.2	3.0
20	3.0	3.6
24	3.8	4.7
28	4.6	5.4
32	5.6	5.7
36	6.8	6.1
40	8.0	6.5
44	9.2	7.2
48	10.4	7.9
52	10.5	7.9
56	10.7	7.9
60	11.0	7.9
64	12.0	7.9
68	14.0	9.0
72	16.0	10.0

3.4.3 Types II and IV trestle and extension ladders.

3.4.3.1 Compression. Ladders shall withstand load test without failure or permanent deformation when tested in accordance with 4.4.3.1.

3.4.3.2 Side rail bending. The side rails shall show no failure or permanent deformation when tested in accordance with 4.4.3.2.

3.4.3.3 Stability. Side rails shall not lift from the floor when tested in accordance with 4.4.3.3.

3.5 Identification and safety marking. Ladders shall be marked with the name or trademark of the manufacturer and the month and year of manufacture. Marking shall also include the following notices in 1/8 inch high letters located on the outside of one of the base side rails:

CAUTION: Metal ladders shall not be used where contact may be made with electrical circuits.

SIZE: Maximum working length____ feet, total length of sections___feet.

3.6 Workmanship.

3.6.1 Exposed metal surfaces. All exposed metal surfaces shall be free from sharp edges in excess of 1/32 inch, burrs in excess of 1/64 inch high, safety hazards and structural defects.

3.6.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have all burrs in excess of 1/32 inch high removed. Flat washers, lock washers, or lock nuts shall be in accordance with good commercial practice. All bolts, nuts, and screws shall be tight or shall permit smooth functioning of component parts when used in pivotally connected joints. A minimum of 1-1/2 threads shall be exposed beyond the nut.

3.6.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have all burrs in excess of 1/32 inch removed. Rivet heads when not countersunk or flattened shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with rivet holes, and in contact with the surface of the member such that for non-pivotable connections a 0.005 inch feeler gauge cannot be fully inserted between the rivet head and the surface of the member or between two members attached together by the rivet.

3.6.4 Welds, appearance. All welds shall be free from undercuts, cracks, and closely spaced in-line surface porosity.

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3.6.5 Rung joints. Rung joints shall be made in accordance with good commercial practice. Where the rung joint itself is beaded or flared to form the connecting means between the rung and the side-rail, the rung shall not show any visible sign of splitting in the bend areas or non-uniform forming of the bead or flare. The bead or flare shall fit up tight against the side rail such that a 0.005 inch feeler gauge shall not fit between the side-rail and the bead or flare.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all ladders of the same type, class, grade, kind, and size offered for delivery at the same time.

4.2.2 Dimensional and visual examination. Sampling for these characteristics shall be in accordance with level S-4 in MIL-STD-105.

4.2.3 Sampling for tests. Ladders shall be sampled in accordance with level S-1 in MIL-STD-105 for tests.

4.3 Examination.

4.3.1 Visual. Ladders shall be examined for defects listed in table VII. The Acceptable Quality Levels (AQL's) in MIL-STD-105 shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

TABLE VII. Classification of defects, end item

Defects	Major	Minor
Type, class, grade, kind, and size not as specified.	X	
Caps not provided.		X
Length and width not within tolerances.	X	
Spacing between rungs or steps not within tolerances.	X	
Overlap not within tolerances, for extension ladders.	X	
Spread of base not within tolerances or less than minimum.	X	
Spreaders not as specified.	X	
Steps or rungs not properly reinforced.	X	
Rungs or steps not skid resistant.	X	
Rungs not securely attached or beaded or flared joint spilt.	X	
Rope and pulley not securely attached.	X	
Rope and pulley incorrect size.	X	
Safety shoes missing, when specified.	X	
Safety shoes not securely attached, when specified.	X	
Side rails twisted or warped.	X	
Parts missing.	X	
Leveling devices missing, when specified.	X	
Hooks missing when specified.	X	
Marking missing.		X
Extension locks not operating correctly.	X	
Binding or excessive free play in movable parts.	X	
Cracks, undercuts, or in-line porosity in welds.	X	
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviceability.		X

4.3.2 Dimensional.

4.3.2.1 Single ladder. Determine size of type I ladders by measuring the overall of a side rail. Exclude any permanently attached foot or safety shoe and the cap.

4.3.2.2 Extension ladder. Determine size of type II ladders by measuring the overall length of one of the side rails of each section and adding these together. Exclude any permanently attached foot or safety shoe and the cap.

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4.3.2.3 Trestle ladders. Determine size of type III ladders by measuring the overall length of one of the side rails. Include any permanently attached foot or safety shoe.

4.3.2.4 Extension trestle ladders. Determine size of type IV ladders by measuring the length of the trestle ladder base along the front edge of the side rail, including the shoe plus the allowable extended length of the extension section, measured along its side rail.

4.3.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VIII. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE VIII. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, distortion.

4.4 Test methods.

4.4.1 Types I, II, III and IV.

4.4.1.1 Hardware, leveling device, and hooks. The test unit shall be either a full size ladder or a short section of sufficient length for test purposes. Short test units shall consist of portions of the base and fly sections of the ladder with all the hardware and any specified fittings attached. When a full size ladder is used extend the fly section a minimum of one rung beyond the minimum working length of the test unit. Place the unit in a vertical position and apply the equally distributed test load in a downward direction to the ends of the side rails of the upper portion of the test unit. The test loads for the ladders shall be in accordance with table IX. (See 3.1.5, 3.1.7, 3.1.8, 3.1.9, and 3.2.6.)

TABLE IX. Test loads, hardware, leveling device, and hooks

Grade	Test load, minimum
	pounds
A	970
B	875
C	775

4.4.1.2 Rungs.

4.4.1.2.1 Bending strength. Use either a single, full-size-section of a ladder or a three-rung section taken from the maximum width portion of a like ladder having the same rung cross-section and rung joint. Set the test unit vertically or at a working angle of 75-1/2 degrees to the horizontal. Apply a distributed load of 800 pounds for 1-minute on a 3-1/2 inch wide block, shaped to the contour of the rung and resting on the center of the widest rung (see 3.4.1.1.1). Also apply test load to the widest rung without bracing. When a 3-foot test sample is used, apply the test load to the center rung.

4.4.1.2.2 Rung to side rail shear strength. Use either a single, full-size-section of a ladder or a three-rung test section taken from a like ladder having the same rung cross section and rung joint. Set the test unit vertically or at a working angle of 75-1/2 degrees to the horizontal. Apply a distributed load of 800 pounds for 1-minute on a 3-1/2 inch wide block shaped to the contour of the rung and resting on the center rung as near the side rail as possible (see 3.4.1.1.2). Apply the test to the rung joint with the least bracing. When a 3-rung test unit is used apply the test to the center rung.

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4.4.1.2.3 Torque. Use either a single-section or a short ladder section comprising at least one rung and two side rails. Apply a torque in accordance with table X to a 3-1/2 inch width at the center of the rung using a bar that does not damage the rung and having a 30 inch long moment arm. Apply the torque first in a clockwise then in a counter-clockwise direction for 10 alternate load cycles (see 3.4.1.1.3).

TABLE X. Rung torque

Grade	Torque, minimum	Load, minimum
	inch-pounds	pounds
A	900	30
B	810	27
C	720	24

4.4.1.2.4 Side rail cantilever bending. Use a straight ladder for type I, the base section for type II, and one leg of the base for types III and IV. Place types I and II on edge with the rungs in a vertical plane. Place types III and IV ladders in an open position on their side with the rungs in a vertical plane. Clamp the lower side-rails to a horizontal support and apply the test load in accordance with table XI for a least 1-minute to the extreme bottom edge of one of the upper side rails (see 3.4.1.2). Remove any safety shoes or spikes affixed to the test section before the test is conducted. Apply the test load to a 2-inch-wide block across the side rail.

TABLE XI. Side rail cantilever bending loads

Grade	Test load, minimum
	pounds
A	200
B	150
C	100

4.4.2 Types I and II, straight and extension ladders.

4.4.2.1 Vinyl caps. Mount a cap or a convenient size specimen in a binocular microscope and examine at 100 to 150X using reflected light. Presence of resin or other descrete particles shall be cause for rejection (see 3.2.7).

4.4.2.2 Horizontal bending. Place the straight ladder or the extension ladder opened to its full working length in a flat horizontal position supported 6 inches in from both sets of side rail ends. Preload the ladder to 75 percent of the test load of table XII for a period of 1-minute in order to set the ladder, unload and then reload the ladder in accordance with table XII. Apply the test loads equally to the side rails at the center of the ladder by means of a 3-1/2 inch wide beam. (See 3.4.2.1).

TABLE XII. Horizontal bending load

Grade	Test load, minimum 1/
	pounds
A	250
B	225
C	200

1/ This test is based on the rated load capacity of the ladder being placed on the ladder with the ladder set at 75-1/2 degrees to the ground. With the rated load of the ladder on the center rung, the component of this weight at right angles to the ladder will be equal to 1/4 the test load. The test load incorproates the required safety factor of 4.

4.4.2.3 Side sway. Place the straight ladder or one section of the extension ladder on edge resting on level supports located 6-inches in from each end. The side rails will be in the horizontal plane and the rungs will be in the vertical plane. Apply a preload of 30 pounds for a period of 1-minute in order to set the ladder, unload and then reload the ladder in accordance with table XIII. Apply the test loads to the center of the span on a 3-1/2 inch wide area of the bottom side-rail (see 3.4.2.2).

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4.4.2.4 Deflection (see 3.4.2.3).

4.4.2.4.1 Deflection test. Place the straight ladder or the extension ladder opened to its full working length in a flat, horizontal position supported 6 inches in from both sets of side rail ends. Preload the ladder with 30 pounds suspended from one of the side rails on a 3-1/2 inch wide area midway between the supports, hold for 1-minute to set the ladder, unload and then reload the ladder with a test load in accordance with table XIII.

TABLE XIII. Deflection load

Grade	Test load, minimum
	pounds
A	60
B	55
C	50

4.4.2.4.2 Measurement and calculation. Determine deflections in table VI by measuring at midpoint between supports, the distance from the outside edges of both rails to the floor or other reference surface both before and after applying the test load in table XIII to one rail of the ladder. Repeat loading the other side of the ladder. Calculate the angle between the load and unloaded rails, and the horizontal as follows:

$$\alpha = \arcsin \frac{d}{w}$$

where

α = Angle between the loaded and unloaded rails and the horizontal
d = Difference in deflection
w = Outside ladder width

4.4.3 Types III and IV, trestle and extension trestle ladders.

4.4.3.1 Compression. Place the ladder to be tested upright in the open position. Apply a uniformly distributed test load, in accordance with table XIV, for at least 1-minute to a 3-1/2 inch wide block centered over the top rungs of type III ladders or the top rung in the extension section of type IV ladders (see 3.4.3.1).

TABLE XIV. Compression load

Grade	Load, minimum
	pounds
A	1,000
B	900
C	800

4.4.3.2 Side-rail bending. Place the ladder to be tested upright in the open position. Apply the test load, in accordance with table XIV, for at least 1-minute to a 3-1/2 inch block with its underside shaped to the contour of the rung. Position the block on the center of the middle rung of type III ladders and the base section of type IV ladders (see 3.4.3.2).

4.4.3.3 Stability. Set the ladder to be tested upright, in the open position, on a level floor (see 3.4.3.3).

4.4.3.3.1 Face stability. Apply a uniformly distributed test load, in accordance with table XV, to the top rungs of type III ladders or the top rung of the extension section of type IV ladders, then apply tension horizontal to the vertical load, in accordance with table XV, to the same rung or rungs for at least 1/2 minute.

4.4.3.3.2 Side stability. Apply a uniformly distributed test load, in accordance with table XV, to the top rungs of type III ladders or the top rung of the extension section of type IV ladders, then apply tension horizontal to the vertical load, in accordance with table XV, to the same rung or rungs for at least 1/2 minute.

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TABLE XV. Stability loads, pounds

Grade	Load, minimum at top rungs or rung	Face pulling force, minimum	Side pulling force, minimum
A	250	80	25
B	225	75	20
C	200	70	15

4.5 Rejection. A lot shall be rejected for failure to comply with any of the specified requirements when tested in accordance with 4.4. Lots rejected for failure may be resubmitted for test provided the lots have been reworked, as necessary, to correct the deficiencies or to remove the nonconforming matertial.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type II, class a or b. Each type II, class a or b ladder shall have the extension sections combined to the minimum length. The sections shall be secured together with tape conforming to PPP-T-97, type II, not less than 1 inch wide or with soft annealed wire, or strapping conforming to QQ-S-781, type I, class B, grade 2 or PPP-S-769, type II. Corner protection pads of wood or other suitable material shall be provided under the steel strapping.

5.1.1.2 Type III or IV. Each type III or IV ladder shall be folded with extension section when applicable and secured with tape conforming to PPP-T-97, type II, not less than 1 inch wide or with soft annealed wire or strapping conforming to QQ-S-781, type I, class B, grade 2 or PPP-S-760, type II. Corner protection pads of wood or other suitable material shall provided under the steel strapping.

5.1.2 Level C. The ladders shall be packaged to afford protection against corrosion or damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), the ladders shall be packed individually. When specified (see 6.2) ladders of the same type, class and grade in quantities as specified (see 6.2) shall be packed in a crate conforming to PPP-C-650, type II, style A. Waterproofing of the ladders shall be in accordance with appendix to the crate specification. The gross weight shall not exceed the weight limitation of the crate specification.

5.2.2 Level B. Each ladder combined to the minimum length with extensions when applicable shall be secured with tape conforming to PPP-T-97, type II, not less than 1 inch wide or with soft annealed wire, or strapping conforming to QQ-S-781, type I, class A, grade 2 or PPP-S-760, type II or III. Corner protection pads of wood or other suitable material shall be provided under the steel strapping. Unless otherwise specified (see 6.2), the ladders shall be packed individually.

5.2.3 Level C. The ladders shall be packed to insure carrier acceptance and safe delivery at destination complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the ladders and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the ladders and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Grade A, types I, II, III and IV. Heavy duty ladders are intended for industrial use such as in construction or industrial maintenance.

6.1.2 Grade B, types I, II, III and IV. Medium duty ladders are intended for light industrial or commercial use such as in office or where heavier loads are not commonly associated with the use of these ladders.

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6.1.3 Grade C, types I and II. Light duty ladders are intended for household use.

6.2 Ordering data. Purchasers should exercise any desired option offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, grade, and kind (see 1.2.1 and 6.3).
(c) Safety shoes (see 1.2.3 and 3.1.8).
(d) Size (see 1.2.2).
(e) Leveling device, when required (see 3.1.9).
(f) Hooks, when required (see 3.2.6).
(g) Size of lot, if different from 4.2.1.
(h) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(i) When ice picks are required in lieu of or in addition to safety shoes (see 3.1.8.4 and 3.1.9).
(j) Quantity of ladders packed in crate (see 5.2.2).
(k) Weight per foot of ladders when other than that normally supplied by the manufacturer (see 3.2.8 and 3.3.5).

6.3 Trestle ladders. Types III and IV are not available for grade c, household use. Both types of trestle ladders are only available in aluminum alloy, kind 1.

6.4 FOR CIVIL AGENCY PROCUREMENTS. There are no level B packaging requirements for civil agency procurements.

6.5 Safety precautions.

6.5.1 Pitch. Portable non-self-supporting ladders should be erected at a pitch of 75-1/2 degrees for maximum balance and strength. A simple rule for setting up a ladder at the proper angle is to place the base a distance from the wall equal to 1/4 the working length of the ladder.

6.5.2 Fastening together. Ladders should not be tied or fastened together to provide longer sections, unless they are already equipped with the necessary hardware fittings.

6.5.3 Improper use. Ladders should not be used as a brace, skid, guy or gin pole, gangway, or for other uses than that for which they are intended.

6.5.4 Electrical hazards. Proper safety measures should be used when metal ladders are used in areas containing electric circuits to prevent short circuits or electrical shocks. Metal ladders are not recommended for electrical maintenance work.

6.5.5 Loading. Ladders covered by this specification are designed as a one-man working ladder including any material supported, based on a 250 pound load capacity for heavy duty ladders, a 225 pound load capacity for medium duty ladders, and a 200 pound load capacity for light duty ladders.

6.5.6 Ladders for household use. While grade C ladders are intended for household use, consideration should be given to grade B ladders for this use.

DOD coordination has been waived.

Preparing activity:
GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies and other documents referenced herein. Price 15 cents each.

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment, detach, complete, and mail to General Services Adm., FSS (FMSH), Washington, D.C. 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

RR-L-91D Ladders, Straight, Extension, and Trestle

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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